

PCB assembly: Control Board

Assembly	PCB assembly	Qty per chassis	1
Internal P/N	HW-C2-PCB-CTRL-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	1	524265-E	305-524265-E
[]	8	971350581	710-971350581
[]	2	ACT45B-101-2P-TL003	810-ACT45B1012PTL003
[]	3	B2B-XH-AM_LF__SN_	306-B2B-XH-AMLFNSNP
[]	36	100nF 100V	80-C1210C104K1RAC
[]	21	10nF 500V	187-CL32B103KGFNNNE
[]	7	10uF 25V	187-CL32A106KAULNNE
[]	8	1uF 50V	187-CL32B105KBHNNNE
[]	2	2.2uF 1210 100v	80-C1210C225J1RAUTO
[]	4	22pF 50V	80-C1210C220J5G
[]	2	8mhz hc49-4h crystal	710-830003156B
[]	1	LTST-C230CKT	859-LTST-C230CKT
[]	1	LTST-C230GKT	859-LTST-C230GKT
[]	1	STM32H723ZGT6	511-STM32H723ZGT6
[]	1	STM32G474RCT3	511-STM32G474RCT3
[]	1	TSW-110-21-L-D-LL	200-TSW11021LDLL
[]	1	TPS389006ADJRTER	595-TPS389006ADJRTER
[]	27	10k 1210	603-AC1210JR-0710KL
[]	2	120 1210	603-AC1210FR-07120RL
[]	9	1k 1210	603-RC1210FR-071KL
[]	16	20k 1210 resistor	667-ERJ-P14F2002U
[]	16	Thick Film Resistors - SMD 1/2watt 249Kohms 1%	71-CRCW1210-249K-E3
[]	1	3k 1210 1%	603-RC1210FR-073KL
[]	2	4.7k 1210	71-CRCW12104K70FKEAH
[]	1	499kohm 1210 resistor	667-ERJ-P14F4993U
[]	2	560ohm 1210 resistor	667-ERJ-P14F5600U
[]	1	RD7-12S033R	737-RD7-12S033R
[]	3	RKE-1205S_H	919-RKE-1205S/H
[]	4	22-04-1041	538-22-04-1041
[]	3	MM3Z3V0ST1G	863-MM3Z3V0ST1G
[]	4	SZMMSZ4678T1G	863-SZMMSZ4678T1G
[]	1	MAX22530AWE+	700-MAX22530AWE+
[]	4	ISO1042BDWVR	595-ISO1042BDWVR

☐	Qty	Description	Part Number
☐	1	CY15B102Q-SXET	727-CY15B102Q-SXET
☐	1	ISO1212DBQ	595-ISO1212DBQ
☐	2	SSW-103-01-L-D	200-SSW10301LD
☐	2	SSW-108-01-L-D	200-SSW10801LD
☐	1	SSW-110-01-F-D	SSW-110-01-F-D
☐	1	STS41A-AWLB-R3	403-STS41A-AWLB-R3
☐	5	Testpoint TH, RED	534-5000
☐	2	BTS462TATMA1	726-BTS462TATMA1
☐	1	LT3080IQ#PBF	584-LT3080IQ#PBF
☐	2	TS40-66-50-R-260-SMT-TR	179-TS406650R260S

Inspection

☐	Check
☐	Visual inspection — no damage, contamination, or solder splash
☐	Solder joints — wetting, no bridges, no tombstoning
☐	IC / connector orientation (pin 1 marks match silkscreen)
☐	Continuity: no shorts across power rails (12 V, 5 V, 3.3 V)
☐	ESD precautions followed during assembly

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

PCB assembly: DC Bus Capacitor Board

Assembly	PCB assembly	Qty per chassis	1
Internal P/N	HW-C2-PCB-DCBUSCAP-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	8	9774030592R	710-9774030592R
[]	60	UCS2D331MHD	647-UCS2D331MHD

Inspection

[]	Check
[]	Visual inspection — no damage, contamination, or solder splash
[]	Solder joints — wetting, no bridges, no tombstoning
[]	IC / connector orientation (pin 1 marks match silkscreen)
[]	Continuity: no shorts across power rails (12 V, 5 V, 3.3 V)
[]	ESD precautions followed during assembly

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

PCB assembly: Filter Board

Assembly	PCB assembly	Qty per chassis	1
Internal P/N	HW-C2-PCB-DCFLT-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	1	524265-E	305-524265-E
[]	12	B58031U9254M062	871-B58031U9254M062
[]	6	MKP1848S61010JY5B	75-MKP1848S61010JY5B
[]	18	VY1222M47Y5UQ6TV0	72-VY1222M47Y5UQ6TV0

Inspection

[]	Check
[]	Visual inspection — no damage, contamination, or solder splash
[]	Solder joints — wetting, no bridges, no tombstoning
[]	IC / connector orientation (pin 1 marks match silkscreen)
[]	Continuity: no shorts across power rails (12 V, 5 V, 3.3 V)
[]	ESD precautions followed during assembly

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

PCB assembly: Gate Driver Board

Assembly	PCB assembly	Qty per chassis	1
Internal P/N	HW-C2-PCB-GD-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	3	524265-E	305-524265-E
[]	8	971150581	710-971150581
[]	32	100nF 100V	80-C1210C104K1RAC
[]	30	10uF 25V	187-CL32A106KAULNNE
[]	6	220pF 100V C0G 1210	80-C1210C221J2G
[]	8	UCM1H101MCL1GS	647-UCM1H101MCL1GS
[]	6	STTH112A	511-STTH112A
[]	1	LTST-C230CKT	859-LTST-C230CKT
[]	1	LTST-C230GKT	859-LTST-C230GKT
[]	6	MGJ2D121509MPC-R7	580-MGJ2D121509MPCR7
[]	1	SSW-110-01-F-D	200-SSW11001FD
[]	2	120 1210	603-AC1210FR-07120RL
[]	8	1k 1210	603-RC1210FR-071KL
[]	12	2.7 ohm 1210	667-ERJ-14BQF2R7U
[]	1	220 1210	603-RC1210FR-07220RL
[]	2	3k 1210 1%	603-RC1210FR-073KL
[]	1	4.7k 1210	71-CRCW12104K70FKEAH
[]	2	MM3Z3V0ST1G	863-MM3Z3V0ST1G
[]	12	BAT54WS-7-F	621-BAT54WS-F
[]	6	NCV57100DWR2G	863-NCV57100DWR2G
[]	16	Testpoint TH, RED	534-5000
[]	2	BTS462TATMA1	726-BTS462TATMA1

Inspection

[]	Check
[]	Visual inspection — no damage, contamination, or solder splash
[]	Solder joints — wetting, no bridges, no tombstoning
[]	IC / connector orientation (pin 1 marks match silkscreen)
[]	Continuity: no shorts across power rails (12 V, 5 V, 3.3 V)
[]	ESD precautions followed during assembly

Assembled by:		Date:	
Inspected by:		Date:	

Signature:		Signature:	
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PCB assembly: IO Board

Assembly	PCB assembly	Qty per chassis	1
Internal P/N	HW-C2-PCB-IO-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	1	776231-1	571-776231-1
[]	8	971250581	710-971250581
[]	2	AWHSH105D00G	523-AWHSH105D00G
[]	2	100nF 100V	80-C1210C104K1RAC
[]	5	UCM1H101MCL1GS	647-UCM1H101MCL1GS
[]	2	S3N	512-S3N
[]	1	EC7BW-110S12	418-EC7BW-110S12
[]	1	MF-RHT200_32-2	652-MF-RHT200/32-2
[]	1	DRDNB21D-7	621-DRDNB21D-7
[]	1	SSW-104-01-L-D	200-SSW10401LD
[]	1	BTS462TATMA1	726-BTS462TATMA1
[]	2	TSW-103-18-G-D	200-TSW10318GD
[]	2	TSW-108-18-F-D	200-TSW10818FD
[]	1	TSW-110-18-T-D	200-TSW11018TD
[]	1	UJ2-B-HR-G-V	179-UJ2-B-HR-G-V

Inspection

[]	Check
[]	Visual inspection — no damage, contamination, or solder splash
[]	Solder joints — wetting, no bridges, no tombstoning
[]	IC / connector orientation (pin 1 marks match silkscreen)
[]	Continuity: no shorts across power rails (12 V, 5 V, 3.3 V)
[]	ESD precautions followed during assembly

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

chassis hardware: Mechanical

Assembly	chassis hardware	Qty per chassis	1
Internal P/N	—	Revision	A

Parts

☐	Qty	Description	Part Number
☐	1	3M156058-ND	3M156058-ND
☐	8	90128A232	90128A232
☐	5	90225A102	90225A102
☐	18	91292A134	91292A134
☐	4	91294A220	91294A220
☐	6	91502A180	91502A180
☐	6	91502A183	91502A183
☐	6	94669A190	94669A190
☐	6	94669A196	94669A196
☐	6	94669A199	94669A199
☐	6	94669A202	94669A202
☐	25	98044A224	98044A224
☐	3	CM600DY-24T	CM600DY-24T

Inspection

☐	Check
☐	Correct part and quantity per line
☐	Fastener torque per specification
☐	Threadlocker applied where required

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

Capacitor Temp Sense Wiring Harness

Assembly	wiring harness	Qty per chassis	1
Internal P/N	HW-C2-WH-TSBCM-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	1	455-2266-ND	306-XHP-2
[]	1	NXFT15XH103FEAB021	81-NXFT15XH103FEAB21
[]	2	SXH-001T-P0.6	306-SXH-001T-P0.6

Inspection

[]	Check
[]	Wire gauges and colors match the schematic
[]	Crimp quality — pull test one crimp per connector
[]	Pin-to-pin continuity against the schematic
[]	Connector keying / locking features engaged
[]	Length and routing match the harness drawing

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

Current Sensor Wiring Harness

Assembly	wiring harness	Qty per chassis	4
Internal P/N	HW-C2-WH-CS-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	8	0022011042	0022011042
[]	32	08-50-0113	08-50-0113
[]	4	LA37S600S05KM	838-LA37S600S05KM

Inspection

[]	Check
[]	Wire gauges and colors match the schematic
[]	Crimp quality — pull test one crimp per connector
[]	Pin-to-pin continuity against the schematic
[]	Connector keying / locking features engaged
[]	Length and routing match the harness drawing

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

Gate Drive Wiring Harness

Assembly	wiring harness	Qty per chassis	1
Internal P/N	HW-C2-WH-GDVS-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	5	544959-E	544959-E
[]	25	464762-E	305-464762
[]	12	FDFNYD1-110(5)	161-FDFNYD1-1105

Inspection

[]	Check
[]	Wire gauges and colors match the schematic
[]	Crimp quality — pull test one crimp per connector
[]	Pin-to-pin continuity against the schematic
[]	Connector keying / locking features engaged
[]	Length and routing match the harness drawing

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

IGBT Temperature Wiring Harness

Assembly	wiring harness	Qty per chassis	2
Internal P/N	HW-C2-WH-TSIGBT-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	2	455-2266-ND	306-XHP-2
[]	2	RTS103C1R2M6L201	527-RTS103C1R2M6L201
[]	4	SXH-001T-P0.6	306-SXH-001T-P0.6

Inspection

[]	Check
[]	Wire gauges and colors match the schematic
[]	Crimp quality — pull test one crimp per connector
[]	Pin-to-pin continuity against the schematic
[]	Connector keying / locking features engaged
[]	Length and routing match the harness drawing

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

fabricated part: Fab

Assembly	fabricated part	Qty per chassis	1
Internal P/N	HW-C2-PLT-BSP-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	1	HW-C2-BSP-A	HW-C2-BSP-A

Inspection

[]	Check
[]	Dimensions match the drawing / STEP model
[]	Edges deburred; finish as specified in info.txt
[]	Flatness / bend angles within tolerance
[]	Threads and holes checked with mating hardware

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

fabricated part: Fab

Assembly	fabricated part	Qty per chassis	1
Internal P/N	HW-C2-PLT-CHSP-A	Revision	A

Parts

☐	Qty	Description	Part Number
☐	1	HW-C2-CHSP-A	HW-C2-CHSP-A

Inspection

☐	Check
☐	Dimensions match the drawing / STEP model
☐	Edges deburred; finish as specified in info.txt
☐	Flatness / bend angles within tolerance
☐	Threads and holes checked with mating hardware

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

fabricated part: Fab

Assembly	fabricated part	Qty per chassis	2
Internal P/N	HW-C2-BB-DCLBB-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	2	HW-C2-DCLBB-A	HW-C2-DCLBB-A

Inspection

[]	Check
[]	Dimensions match the drawing / STEP model
[]	Edges deburred; finish as specified in info.txt
[]	Flatness / bend angles within tolerance
[]	Threads and holes checked with mating hardware

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	

fabricated part: Fab

Assembly	fabricated part	Qty per chassis	3
Internal P/N	HW-C2-BB-PBB-A	Revision	A

Parts

[]	Qty	Description	Part Number
[]	3	HW-C2-PBB-A	HW-C2-PBB-A

Inspection

[]	Check
[]	Dimensions match the drawing / STEP model
[]	Edges deburred; finish as specified in info.txt
[]	Flatness / bend angles within tolerance
[]	Threads and holes checked with mating hardware

Assembled by:		Date:	
Inspected by:		Date:	
Signature:		Signature:	